

COGNITIVE ABILITY AND BIDDING BEHAVIOR IN SECOND PRICE AUCTIONS: AN EXPERIMENTAL STUDY

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This paper examines what connection, if any, there is between cognitive ability and bidding strategy in second price auctions. Despite truthful revelations being a weakly dominant strategy, previous experiments have consistently observed overbidding, which makes the use of such auctions for inferring homegrown values problematic. Examining the effect of cognitive ability is important, as it may help identify when one can reliably recover values from observed bids. The results indicate that more cognitively able subjects behave in closer accordance with theory, and that cognitive ability partially explains heterogeneity in bidding behavior. Our results suggest that considering subjects' cognitive ability in homegrown valuation studies can help identify the true underlying demand conditions.

Key words: Bid deviation, cognitive ability, homegrown valuation, overbidding, second price auction.

JEL codes: C91, C92.

The private value second price sealed bid auction is a popular tool for applied economists to elicit people's willingness to pay (WTP) and/or willingness to accept (WTA) for new products, product attributes, new policies, and so on. Findings from valuation research are used not only for marketing and business applications but also for public policy and welfare analysis. The second price auction (SPA) is considered demand revealing because truthful bidding is the weakly dominant strategy. This theoretical prediction means that bidding behavior should not be influenced by information about rival bidders, a bidder's risk attitude, and so forth. However, this auction mechanism may not always provide an

accurate measure of value given behavioral anomalies observed in controlled laboratory experiments. One consistent finding in experimental auction studies is that subjects tend to deviate from rational behavior and exhibit a pattern of overbidding in SPAs (Kagel, Harstad, and Levin 1987; Kagel and Levin 1993; Rutström 1998; Harstad 2000; Aseff 2004; Andreoni, Che, and Kim 2007; Cooper and Fang 2008; Drichoutis, Lusk, and Nayga 2015; Georganas, Levin, and McGee 2017).

Several experimental studies have tried to explain this phenomenon in SPAs. Kagel, Harstad, and Levin (1987) conjectured that subjects submit a higher bid due to the illusion that bidding higher improves the probability of winning with no real cost because the highest bidder pays the second highest bid. Alternatively, Morgan, Steiglitz, and Reis (2003) explained bid deviations as being based on behavioral motives such as "spite." They suggested that subjects bid more aggressively in a SPA because the profit earned by a rival bidder could be reduced by a losing bidder's own bid. The overbidding behavior can be also explained by a "joy of winning." Subjects would tend to bid higher than their own values

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if they derive extra utility from winning the auction. Cooper and Fang (2008) found that small and medium overbidding is consistent with a “(modified) joy of winning” hypothesis, but large overbidding is more consistent with the “spite” hypothesis.

Understanding bid deviation and overbidding in the SPA is important not only for explaining individuals’ apparently irrational behavior in a strategic game but also for providing clarity as to when and how bids should be interpreted when trying to elicit home-grown WTP or WTA valuations. Given that cognitive ability can vary considerably across individuals, it is important to know the role of cognition to obtain better predictions of market behavior when using the SPA. In this paper, we investigate how individuals’ cognitive ability influences bid deviations in SPAs. Our study is motivated by previous work by Kagel, Harstad, and Levin (1987) and Ausubel (2004), who argued that the difficulty of understanding the SPA could lead to the prevalence and persistence of overbidding in the SPA as compared to the ascending price English auction, even though both auction mechanisms are strategically equivalent. Specifically, they explained that subjects in English auctions are more likely to play the dominant strategy because they are simpler to understand than the SPA (Kagel, Harstad, and Levin 1987; Ausubel 2004). Using home-grown value auctions, Rutström (1998) confirmed that subjects in the SPA tend to submit higher bids than those in the English auction and claimed that subjects in the SPA cannot perceive an expected loss from overbidding due to a lack of the public information available in English auctions.

More recently, Li (2017) suggested that the bid deviation observed in SPAs is due to the fact that it is not an obviously strategy-proof (OSP) mechanism, where an OSP mechanism is described as one a cognitively limited agent can recognize as the weakly dominant strategy. Here, a cognitively limited subject is a person who cannot fully use contingent reasoning to break the state space into separate contingencies and compare strategies one contingency at a time (Li 2017). The OSP mechanism does not require contingent reasoning, so the cognitively limited subject is able to recognize the dominant strategy. Li (2017) proved that ascending price English auctions are obviously strategy proof, whereas SPAs are strategy proof but not obviously strategy proof. Hence, given the well-defined positive

relationship between cognitive ability and reasoning ability, subjects with higher cognitive ability will understand the strategic properties of the SPA better than subjects with lower cognitive ability, and thus, they are less likely to deviate from rational bidding behavior in SPAs. Given this conjecture, our study tests whether individuals’ cognitive ability can explain bid deviation in a SPA. Although previous studies have examined the relationship between cognitive ability and economic behavior, no other study has examined how cognitive ability affects bidding in SPAs. Our study also answers the question of how subjects’ aggregate bidding behavior depends on both their own cognitive ability and their beliefs about the ability of other subjects in the auction. In SPAs, information about the type of other bidders should not matter, as the dominant strategy is to bid one’s own value regardless of what other subjects do. But if information about cognitive types makes a difference, then this would be an interesting and important point to understand.

Our study is composed of two experiments. In the first experiment (Experiment 1), in order to investigate the relationship between bidding behavior in the SPA and cognitive ability, we first measured *student* subjects’ cognitive abilities using a nonverbal Raven’s Standard Progressive Matrices (RSPM) test (Gray and Thompson 2004). We then classified subjects into two groups (i.e., a high cognitive ability group and low cognitive ability group) based on their relative RSPM test performance. Each group then participated in a series of second price induced value auctions. In the second experiment (Experiment 2), we recruited subjects from the *general population* to check the robustness of our results. We specifically focused on the role of cognitive ability on individual bidding behavior in the second experiment.

Our results suggest that people with higher cognitive ability are less likely to deviate from the dominant strategy in the SPA, whereas those with lower cognitive ability are more likely to substantially overbid or underbid. This finding provides evidence that more cognitively able individuals behave in closer accordance with theory, which is consistent with the findings from previous studies that cognitive ability is positively related to rational behavior. In addition, our results reflect the importance of cognitive ability in explaining bid deviation in the SPA and can shed light on the critical question in the literature of why

bidders overbid or underbid in SPAs. These are important findings because they imply that in order to minimize behavioral anomalies and make the SPA a valid measurement instrument, researchers should take care in educating subjects about the mechanism, especially those with lower cognitive ability. Overall, our findings provide important evidence that cognitive skills can partially explain heterogeneity in bidding behavior in SPAs. Our results also support the notion that the difficulty of understanding the SPA (especially for cognitively limited agents) can contribute to irrational bidding behavior, which squares well with the fact that the SPA is not an obviously strategy-proof mechanism as defined by Li (2017).

Related Literature

The behavior of people with higher cognitive ability can be different from those with lower cognitive ability in a variety of ways. For example, Dohmen et al. (2010) examined whether risk aversion and impatience are associated with cognitive ability. Unlike a previous study that used hypothetical measures of risk aversion and impatience (Frederick 2005), Dohmen et al. (2010) measured risk aversion and impatience in paid controlled experiments with incentive compatible methods. Dohmen et al. (2010) found that risk aversion and impatience vary with cognitive ability. Specifically, more cognitively able individuals were more likely to take risks, and they were also more patient relative to less cognitively able individuals. Other studies have also shown that high cognitive ability is positively correlated with patience (Frederick 2005; Slonim, Carlson, and Bettinger 2007; Oechssler, Roider, and Schmitz 2009), whereas it is negatively associated with risk aversion (Frederick 2005; Benjamin, Brown, and Shapiro 2013).¹ Smith, McArdle, and Willis (2010) and Agarwal and Mazumder (2013) used secondary data to examine the effects of survey respondents' cognitive ability on their financial behaviors. They found that people with higher cognitive ability are more likely to participate in financial markets and

receive more financial return than people with lower cognitive ability. In addition, cognitive ability is highly correlated with saving behavior and therefore is a good predictor of saving performance (Ballinger et al. 2011).

Our study also builds on the literature connecting individuals' cognitive abilities and strategic behavior. For example, Gill and Prowse (2016) investigated the effects of cognitive ability and character skills on how people learn to play equilibrium in the *p-beauty* contest.² They measured the cognitive ability of subjects using a Raven test, and then they classified a subject into one of three cognitive groups. Each group of three then played the paid *p-beauty* contest. They found that subjects with higher cognitive ability more frequently play equilibrium strategies and also earn more money. Burnham et al. (2009), Brañas-Garza, Garcia-Muñoz, and González (2012), Carpenter, Graham, and Wolf (2013), and Civelli and Deck (2018) found a similar relationship between cognitive ability and performance in the *p-beauty* contest as well. However, these studies, with the exception of Burnham et al. (2009), relied on student subjects. Jones (2014) examined whether cooperation in the repeated prisoner's dilemma games is sensitive to the complexity of available strategies. The results of the incentivized experiment show that cognitive costs associated with strategic complexity influence cooperation in the strategic game, which indicates a positive relationship between cognitive ability and cooperation.

Finally, our study also builds on a strand of the literature investigating the relationship between cognitive ability and the likelihood of exhibiting behavioral biases. Oechssler, Roider, and Schmitz (2009), using an incentivized Internet-based experiment, show that individuals with lower cognitive ability are likely to exhibit higher incidence of the conjunction fallacy and conservatism fallacy as compared to higher cognitive ability people.³

¹ Some studies used hypothetical measures or small stakes monetary choices of risk aversion and patience (e.g., Frederick 2005; Benjamin, Brown, and Shapiro 2013). Other studies relied on student subjects and thus restricted sample heterogeneity (e.g., Slonim, Carlson, and Bettinger 2007; Oechssler, Roider, and Schmitz 2009; Benjamin, Brown, and Shapiro 2013).

² The *p-beauty* contest is a strategic game in which players try to guess which number will be the closest to a fraction p of the average guess of all of the players. While naïve players will guess randomly, more sophisticated players will attempt to best respond to what they believe others will do. Thus, this game is often used to measure the number of iterations of reasoning in which a person engages.

³ The conjunction fallacy is a reasoning failure that occurs when people perceive the probability of a conjunction of events more likely than the probability of one of the constituent events (Tversky and Kahneman 1983). The conservatism fallacy is a bias when people underweigh new evidence and overweigh the base rate (Edwards 1968).

Hoppe and Kusterer (2011) use an incentivized laboratory experiment to examine base rate fallacy and conservatism. They find that individuals with higher cognitive ability are less susceptible to the base rate fallacy⁴ and conservatism bias. Bergman et al. (2010) employ the Becker-DeGroot-Marschak procedure with real money and actual products, and find that greater cognitive ability moderates the anchoring effect. Andersson and Svensson (2008) also tested the hypothesis of a positive correlation between cognitive ability and scale bias in contingent valuation surveys. They measured subjects' cognitive ability and then asked them to state their willingness to pay to reduce mortality risk. Their results indicate that respondents with higher cognitive skills give answers that are less influenced by scale bias.⁵ In summary, the literature suggests that behavioral biases are more pronounced or are more likely among individuals with lower cognitive abilities.

Experiment 1

Experimental Design

As previously mentioned, we used student subjects in Experiment 1. A total of fifteen sessions were conducted, with each session consisting of 10 subjects for a total of 150 subjects.⁶ All sessions were conducted on weekdays between March 2017 and June 2017 in the Behavioral Business Research Laboratories at The University of Arkansas, and each session lasted for approximately ninety minutes. The subjects, a majority of whom were undergraduate students, were recruited from the behavioral lab's database of volunteers. Subjects were paid \$7.50 if they show up on time, and they were paid an extra \$2.50 if they participated in the study. Subjects were also paid on their earnings in the auctions, which averaged \$4.50 ($SD = 3.24$,

$\min = -2.50$, $\max = 12.94$), at the end of the experiment.⁷

Upon arrival, participants signed a consent form and were then seated at a computer visually isolated from the other subjects. Computerized instructions were provided to each subject along with paper copies so that subjects could refer back to the instructions at any point. The instructions were read aloud to aid in comprehension and to ensure that all subjects knew that everyone had received the same instructions. The experiment was comprised of three tasks in the following order: (1) the RSPM, (2) the induced value SPAs, and (3) a questionnaire. The experiment was conducted using the z-Tree software (Fischbacher 2007).

We first measured the cognitive ability of all subjects using the RSPM, which is used for measuring abstract reasoning and is considered a nonverbal estimate of fluid intelligence.⁸ The RSPM consists of five parts, labeled A to E, with twelve questions in each part. The subjects were given three minutes to complete each of the first two parts (i.e., parts A and B) and eight minutes to complete each of the last three parts (i.e., parts C, D, and E). Subjects could move back and forth within each part and change their answers. We did not provide any monetary incentives for each correct answer during the RSPM test to avoid a potential money effect influencing the auction part of the study that followed.⁹ We also provided two example RSPM tasks prior to the actual RSPM. All of the procedures regarding the administration of the RSPM were standard.

We classified a subject as "high cognitive ability" if his/her test score was in the top half of all scores in the session or as "low cognitive ability" if his/her test score was in the bottom

⁴ The base rate fallacy is a tendency that occurs when people ignore the base rate and focus more on new evidence (Kahneman and Tversky 1982).

⁵ The scale bias was defined in the study as the insensitivity of participants' responses to the size of the risk reduction in a valuation survey.

⁶ We addressed the issue of whether our sample size is large enough to detect even small differences in cognitive ability in the Supplementary Material A in the Appendix. We used these sample size calculations as a stopping rule for our experiment.

⁷ At the end of the experiment, on average, subjects were paid \$14.50 ($\min = 7.50$, $\max = 22.94$). A \$7.50 participation payment fee is lab policy and participants could not leave the study with less than this amount. The additional \$2.50 endowment was provided so that any losses (up to \$2.50) could be deducted from this amount so that losses were salient. If a subject lost \$2.50 or more during the study, which five subjects did, he/she was only paid \$7.50. We included a dummy variable for these subjects in the regression analysis and found consistent results with and without this dummy variable. The average salient earnings (exclusive of the show up payment and endowment) for subjects with greater cognitive ability was \$4.36 in the auction, while the average profit for subjects with lower cognitive ability was \$3.57 in the auction. The difference is not statistically significant (t-statistic: -0.86 ; p-value: 0.38).

⁸ Placing a time limit on the Raven test is also considered a measure of intellectual efficiency.

⁹ It is common not to provide monetary incentives during the RSPM test in the psychology and psychometric studies (Gill and Prowse 2016). Furthermore, previous studies observed no significant effect of financial incentives on performance of intelligence test (Borghans, Meijers, and Ter Weel 2008; Gignac 2018).

half of all scores in the session. That is, the ten subjects in the session were split into two groups of five people based on relative performance. To examine whether subjects' bidding behavior depends on their beliefs about the cognitive ability of other members in the group, we informed each subject of their own classification and that of other members in the group. More specifically, subjects in the "high cognitive ability" were informed that everyone in their auction group was in the "top half of all scores of all participants in the room" and subjects in the "low cognitive ability" were informed that everyone in their auction group was in "bottom half of all scores of all participants in the room." We did not reveal subjects' exact RSPM scores until the end of the experiment. The number of subjects in each group was kept constant to eliminate a potential confound of the size of the auction group (see discussion in Drichoutis, Klonaris, and Papoutsis 2017, and Canavari et al. 2019). Controlling for a subject's test score, we checked whether the allocation to the high or low ability group affects subjects' subsequent behavior. We found no evidence that it does (See Supplementary Material B in the Appendix).

Each group then participated in five rounds of a second price induced value auction, where subjects submitted non-hypothetical bids for their private induced values. Prior to the auction, we carefully explained how the second price auction works. Subjects also participated in two practice rounds to help them understand the auction mechanism.¹⁰ In every round of the actual auction, each subject was assigned an induced value for a fictitious good, but no subject ever learned the induced value of another bidder. Subjects were told truthfully that the induced values were drawn from \$0 to \$30. To reduce exogenous variation across sessions and subjects, a single set of induced values was pre-drawn and used for every session. For each and every auction, the induced values were \$5.18, \$11.26, \$17.16, \$23.82, and \$29.12.¹¹ Each subject experienced each of these values in exactly one round of the auction

(out of five), and each value was induced for every auction.¹² Thus, the effective demand curve was constant across all auctions within and between groups.

In the auction, subjects were asked how much they would be willing to pay for a fictitious good.¹³ The subject placing the highest bid bought the good at the second highest bid and then resold it to the experimenter at his or her own induced value at the end of each round. The highest bidder's profit in each round is equal to the difference between his or her induced value and the second highest bid. Profits of subjects who do not purchase the good are zero. Subjects were paid their cumulative earnings from all five rounds at the end of the experiment. Feedback about the highest bid in the auction, the second highest bid, and own or other subjects' earnings were not provided between rounds.

After the auction, subjects were asked to complete a questionnaire containing demographics and character skill questions (i.e., personality traits: openness, conscientiousness, extraversion, agreeableness, and emotional stability; grit: perseverance and passion for long-term goals; and consideration of future of consequences). We measured personality traits using the ten-item assessment of the Big Five (Gosling, Rentfrow, and Swann 2003). We measured grit using the twelve-item Grit Scale (Duckworth et al. 2007) and consideration of future of consequences (CFC) using the twelve-item CFC Scale (Strathman et al. 1994).^{14,15} While these

¹² We randomly chose five induced values from \$0 to \$30 prior to conducting the first session of the experiment. We then used these values for every auction in every session to reduce variation. Given that each subject experienced each of these values in sequence while likely believing that each draw in each auction was independent, a reviewer noted that some readers may view this as deceptive. While each statement given to the subjects was true, we did not try to dispel them of notions they might have held, and they were not informed that the values would be reused and that each subject would rotate through the set of pre-drawn values. Clearly, had the subjects known how values were allocated, they would likely have chosen to behave differently. Not providing any feedback between auctions does help further mask how values were determined. Our procedures are similar to Drichoutis, Lusk, and Nayga (2015), Jacquemet et al. (2013), and Cherry et al. (2004).

¹³ Experimental instructions for student subjects are provided in the Supplementary Material in the Appendix.

¹⁴ While not as well-known as the Big Five, Grit has increasingly been used to explain various behavioral outcomes. The Grit Scale measures "perseverance and passion for long-term goals" (Duckworth et al. 2007). CFC is a survey-based measure of time preference (Strathman et al. 1994). Gill and Prowse (2016) found that these character skills are associated with people's strategic behaviors.

¹⁵ Following the questionnaire, data were also collected for another test of cognitive ability as part of a different research project. That project developed a set of thousands of nonverbal

¹⁰ Feedback about whether the subject is the winner, the amount of the second highest bid, and subject's own profit were provided in the practice auction to help the subjects understand the auction mechanism, but we did not provide any feedback between rounds in the actual paid auction to avoid potential confounds. Corrigan et al. (2012) discourage the use of price feedback between rounds.

¹¹ These five induced values were randomly drawn from a distribution of \$0 to \$30 before the first session. Given the realized draws, the expected price was \$23.82, and the expected profit to the winner was \$5.30.

Table 1. Comparison of Subjects' Characteristics across Cognitive Groups

Variables	All	High cognitive group	Low cognitive group
RSPM***	52.20 (4.52)	55.37 (2.46)	49.02 (3.83)
Age (years)	24.01 (6.79)	23.69 (4.82)	24.32 (8.32)
Gender (1: male, 0: female)	48.00%	49.33%	46.67%
Grade point averages (GPA)***	3.39 (0.51)	3.53 (0.46)	3.25 (0.52)
Employed (1: yes, 0: no)	59.33%	61.33%	57.33%
Big5: Extraversion	8.48 (3.09)	8.72 (3.08)	8.24 (3.10)
Big5: Agreeableness	9.47 (2.47)	9.54 (2.58)	9.39 (2.36)
Big5: Conscientiousness	10.59 (2.69)	10.31 (2.92)	10.88 (2.42)
Big5: Emotional stability	9.54 (2.98)	9.96 (2.92)	9.12 (2.99)
Big5: Openness	10.65 (2.31)	10.73 (2.22)	10.57 (2.41)
Grit	41.23 (7.07)	41.07 (7.35)	41.40 (6.82)
CFC	42.59 (6.37)	42.91 (6.23)	42.27 (6.54)

Note: Each value represents the average of each variable. Standard deviation for each value shown in parentheses. *** denotes the mean difference between the two groups are statistically significant at a 1% level. The Big five personality traits use a seven-point Likert scale to measure responses to individual questions; the Grit scale and CFC use a five-point Likert scale to measure responses to each question.

measures are not of primary interest, we elicited these in order to include them as control variables in our analysis given that each one of them has been found to influence behavioral outcomes (e.g., Grebitus, Lusk, and Nayga Jr. 2013).

Experimental Results

Table 1 compares the characteristics of subjects in the high and low cognitive ability groups. The mean RSPM test score for all subjects was 52.20 (correct out of sixty puzzles), with scores ranging from 37 to 60. The average test score for the higher cognitive group was 55.37 (ranging from 48 to 60), which was about 6.3 higher than the average of the lower cognitive group (mean 49.02, ranging from 37 to 56)¹⁶. The mean difference of the test scores between the two cognitive groups was statistically significant (t-statistic: -12.05 ; $p < 0.001$). We also reject the hypothesis that the variances of the test scores between the two cognitive groups are similar (F-statistic: 2.42; $p < 0.001$)¹⁷. Otherwise, the two cognitive groups have similar observed characteristics. Specifically, the mean differences of demographics and character skills between the two groups are not statistically

significant with the exception of self-report GPA (t-statistic = -3.51 ; $p < 0.001$ for testing equal GPA between groups). This is not surprising because previous studies have found a high correlation between GPA and general cognitive ability (Jensen 1998; Frey and Detterman 2004) while other studies have used GPA as a measure of cognitive ability (Benjamin, Brown, and Shapiro 2013; Chen et al. 2013).¹⁸ That the groups are similar except for cognitive ability suggests that any behavioral differences are driven by that and not other traits.

Overview of Bidding Behavior

The bidding behavior from the experiment is summarized in table 2, where we separate bids into three categories: *perfect demand revealing*, *overbid*, and *underbid*. Results show that on average 29.4% (ranging from 26.7 to 30.7) of bids in the group of subjects with greater cognitive skills are perfectly demand revealing, while only 13.4% (ranging from 10.7 to 14.7) of bids in the lower cognitive group are perfectly demand revealing. The two-sample proportion tests also reject the null hypothesis of equal proportions of perfectly demand revealing bids between two cognitive groups across rounds,¹⁹ indicating that subjects with greater cognitive ability more often bid truthfully.

puzzles as an extension of or substitutes for the Raven test. The result of that study showed that the two tests have similar predictive performance.

¹⁶ The average test score of the higher cognitive group is between the 75th and 90th percentile of the general population, while the average test score of the lower cognitive group is between the 25th and 50th percentile of the general population (Source: Revan Manual in Standard Progressive Matrices).

¹⁷ Detailed summary statistics of the test scores between the groups are reported in Supplementary Material C in the Appendix.

¹⁸ GPA and RPSM scores are positively correlated in our data ($\rho = 0.23$). Based on regression analysis, a 1% change in the Raven test score is associated with a 0.42% change in GPA, and it is statistically significant (p -value = 0.01).

¹⁹ The test results across rounds are as follows: z-statistic: 3.02, p -value: 0.002 in round 1; z-statistic: 2.34, p -value: 0.019 in round 2; z-statistic: 2.34, p -value: 0.019 in round 3; z-statistic: 2.27, p -value: 0.022 in round 4; z-statistic: 1.99, p -value: 0.046 in round 5.

Table 2. Frequency of Bids Relative to Induced Value by Group

			Number			Percentage		
			=	>	<	=	>	<
High cognitive ability	by Round	R1	23	28 (16)	24 (17)	30.7	37.3 (57.1)	32.0 (70.8)
		R2	23	29 (18)	23 (19)	30.7	38.7 (62.1)	30.7 (82.6)
		R3	23	32 (20)	20 (15)	30.7	42.7 (62.5)	26.7 (75.0)
		R4	20	36 (26)	19 (14)	26.7	48.0 (72.2)	25.3 (73.7)
		R5	21	31 (24)	23 (18)	28.0	41.3 (77.4)	30.7 (78.3)
	By value	5.18	25	30 (17)	20 (14)	33.3	40.0 (56.7)	26.7 (70.0)
		11.26	25	25 (14)	25 (21)	33.3	33.3 (56.0)	33.3 (84.0)
		17.16	19	32 (19)	24 (20)	25.3	42.7 (59.4)	32.0 (83.3)
		23.82	22	35 (28)	18 (10)	29.3	46.7 (80.0)	24.0 (55.6)
		29.12	19	34 (26)	22 (18)	25.3	45.3 (76.5)	29.3 (81.8)
Low cognitive ability	By round	R1	8	33 (18)	34 (31)	10.7	44.0 (54.5)	45.3 (91.2)
		R2	11	28 (12)	36 (27)	14.7	37.3 (42.9)	48.0 (75.0)
		R3	11	30 (13)	34 (26)	14.7	40.0 (43.3)	45.3 (76.5)
		R4	9	30 (17)	36 (32)	12.0	40.0 (56.6)	48.0 (88.9)
		R5	11	31 (14)	33 (29)	14.7	41.3 (45.2)	44.0 (87.9)
	By value	5.18	12	29 (7)	34 (26)	16.0	38.7 (24.1)	45.3 (76.4)
		11.26	10	32 (16)	33 (28)	13.3	42.7 (50.0)	44.0 (84.4)
		17.16	9	31 (14)	35 (33)	12.0	41.3 (45.1)	46.7 (94.3)
		23.82	10	30 (18)	35 (27)	13.3	40.0 (60.0)	46.7 (77.1)
		29.12	9	30 (19)	36 (31)	12.0	40.0 (63.3)	48.0 (86.1)

Note: =, >, and < denote bid equal to induced value, bid larger than induced value, and bid smaller than induced value. Entries in parentheses are the number/percentage of individuals whose bids are within 10% of his/her induced value among subjects who overbid or underbid.

The number of subjects who bid higher than induced value is similar in both cognitive groups. However, among subjects who bid over the induced value, subjects in the higher cognitive group are more likely to submit bids within 10% of their induced value (on average, 66.3% of overbids by high types ranging from 57.1 to 77.4) as compared to those in the lower cognitive group (on average, 48.5% of overbids by low types ranging from 42.9 to 56.6), indicating that subjects in the lower cognitive group tend to place larger overbids. More subjects in the lower cognitive group (on average 46.1% of bids) bid lower than the induced value compared to those in the higher cognitive group (on average 29.1% of bids). Among subjects who bid lower than the induced value, on average, 83.9% of bids by low types are within 10% of induced value and 76.1% of bids by high types are within 10% of induced value, indicating that underbidding is typically only a small deviation regardless of cognitive ability.

Interestingly, the percentage of bids within 10% of induced value increased from round 1 (75% of bids) to round 5 (84% of bids) in the higher cognitive group, while no clear pattern was found in the lower cognitive group (76% of bids in round 1 and 72% of bids in round 5 were within 10% of induced value). This suggests that subjects with greater cognitive skills

may be better at learning to submit bids close to the optimum when they have more experience bidding in the second price auction despite not receiving any feedback between auctions. Interestingly, relatively high overbids (bids more than 10% above induced value) in the higher cognitive group significantly vanish with bidding experience²⁰. This finding is consistent with other work on the role of cognitive ability in strategic games – for example, Gill and Prowse (2016) found that subjects with greater cognitive skills choose numbers close to equilibrium in the *p-beauty* contest.

Based on the induced values used in the auction, the percentages of relatively large overbids and underbids (bids above and below 10% of induced value, respectively) in both cognitive groups sharply decrease as their induced values increase, indicating that when subjects have relatively high values for the auctioned item, they tend to bid close to the optimum, which is consistent with the findings from Lusk, Alexander, and Rousu (2007) and

²⁰ Extreme overbids, a bid high enough to cause bankruptcy when a subject is the highest bidder in a group, also decrease with more experience in the higher cognitive group. We found four extreme overbids in round 1 and only one extreme overbid in round 5 in the higher cognitive group. On the other hand, we found six extreme overbids in round 1 and five extreme overbids in round 5 in the lower cognitive group.

Table 3. Absolute Deviations of Bids from Induced Value by Group

			All bid	Overbid	Underbid
High cognitive ability	By round	R1	1.24 (2.84)	2.53 (4.15)	0.92 (1.45)
		R2	1.33 (2.80)	2.32 (3.51)	1.43 (2.76)
		R3	1.95 (7.78)	3.86 (11.62)	1.15 (2.01)
		R4	2.05 (8.10)	3.50 (11.51)	1.45 (1.84)
		R5	1.61 (6.09)	2.97 (9.19)	1.24 (2.07)
	By value	5.18	0.70 (1.51)	1.54 (2.12)	0.32 (0.41)
		11.26	1.70 (8.06)	4.54 (13.67)	0.57 (0.83)
		17.16	1.48 (2.89)	2.41 (3.54)	1.41 (2.63)
		23.82	2.20 (7.83)	3.47 (11.20)	2.44 (2.53)
		29.12	2.09 (6.43)	3.58 (9.20)	1.62 (2.26)
Low cognitive ability	By round	R1	2.10 (6.82)	4.09 (9.96)	0.67 (1.08)
		R2	1.79 (3.38)	3.14 (4.05)	1.28 (2.96)
		R3	3.72 (15.63)	7.34 (24.09)	1.73 (4.21)
		R4	3.49 (16.93)	7.54 (26.42)	0.99 (2.08)
		R5	2.11 (4.35)	3.45 (3.96)	1.55 (5.05)
	By value	5.18	2.89 (16.70)	6.73 (26.65)	0.63 (1.23)
		11.26	1.63 (3.19)	2.83 (4.03)	0.96 (2.27)
		17.16	3.84 (15.62)	8.14 (23.65)	1.01 (2.93)
		23.82	2.29 (4.56)	3.59 (5.53)	1.83 (4.01)
		29.12	2.57 (7.41)	4.36 (10.25)	1.71 (4.87)

Note: Standard deviations are shown in parentheses.

Drichoutis, Lusk, and Nayga (2015). Because relatively high induced values could be associated with a high cost of “misbehavior” (Harrison 1989, 1992), subjects may put more efforts into the tasks. For the higher cognitive group, about 80% and 77% of overbids are within 10% of the induced values when values for the item are \$23.82 and \$29.12, respectively, suggesting that many overbids may be characterized by small mistakes when values for the item are relatively high.²¹ Many subjects with lower cognitive skills tend to bid substantially above induced value (about 76% of overbids are above 10% of the induced value) when they have a low value for the item, and it sharply decreases as induced values increase.

Another way to compare bidding behavior between the two cognitive groups is to look at their bid deviations from induced value. Table 3 reports the mean absolute deviation of revealed bid from induced value, with their standard deviations in parentheses. We also classified the bids as either an overbid or underbid in the table. The results from all bids indicate that on average the deviations of bid from induced values in the higher cognitive group are generally smaller than those in the lower cognitive group. The Mann–Whitney

non-parametric tests reject the null hypothesis that bid deviations from induced value are equally distributed in both cognitive groups (χ^2 -statistic: 12.30, $p < 0.001$), suggesting that subjects in the higher cognitive group tend to submit bids closer to their induced value; that is, they misbid less.

The bid deviations between the two cognitive groups are similar when bids are underbids. The Mann–Whitney test fails to reject the hypothesis that bid deviations are equally distributed in both groups when these are underbids (χ^2 -statistic: 1.80, $p > 0.15$). However, the bid deviations in the lower cognitive group are relatively bigger than those in the higher cognitive group when bids are overbids. The Mann–Whitney test rejects the similar bid distribution between the two cognitive groups (χ^2 -statistic: 9.42, $p < 0.01$), suggesting that subjects with lower cognitive ability tend to submit large overbids. Consistent with previous experimental studies, the magnitude of bid deviation in the overbid case is bigger than that in the underbid case in both groups, resulting in overbidding on average.²²

²¹ We only find one and two extreme overbids, a bid high enough to cause bankruptcy when a subject is the highest bidder in a group, when the induced values are \$23.82 and \$29.12, respectively, while we find seven cases when the induced value is \$5.18.

²² We also test whether bid deviations mainly come from either overbids or underbids based on a regression analysis. The results with and without controls are reported in the Supplementary Material D in the Appendix. The dependent variable is absolute deviations of revealed bids from induced value. We included two dummies – one for overbids and the other for underbids. The results show that the overbids mainly caused the bid deviation in our SPA.

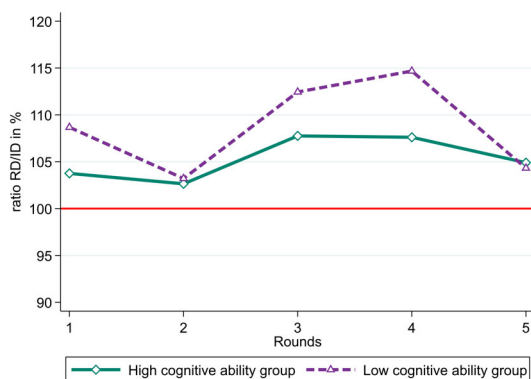


Figure 1. Aggregate behavior by rounds by cognitive ability group

Cognitive Ability and Aggregate Bidding Behavior

Given the importance of aggregate values for policy evaluation and welfare analysis in homegrown value research, we identify the role of cognitive ability (i.e., own cognitive type and information about the type of other bidders) in aggregate bidding behavior in SPAs. We examine aggregate bidding behavior by round in each cognitive group, following the framework of analysis used in Drichoutis, Lusk, and Nayga (2015). We constructed the ratio of aggregate revealed demand over aggregate induced demand in each round. Because the induced values were held constant across auctions, the aggregate induced demand for each group in a given round of the auction is $\$1298.10 = 15 \text{ sessions} \times (\$5.18 + \$11.26 + \$17.16 + \$23.82 + \$29.12)$.

Figure 1 shows the ratio of the revealed demand to induced demand by round for both cognitive groups. The horizontal line at 100% denotes perfect demand revelation. Both cognitive ability groups have similar patterns over the rounds and neither is demand revealing. The ratio in the higher cognitive ability group is always below 110%, but this is not the case for the low cognitive ability group, which also exhibits more variation.

The results generally indicate that regardless of cognitive ability type, in aggregate, subjects tend to overbid in the second price auction, which is consistent with previous findings (Kagel, Harstad, and Levin 1987; Kagel and Levin 1993; Cooper and Fang 2008; Drichoutis, Lusk, and Nayga 2015). However, the pattern of overbidding is somewhat more moderated in the group of subjects with high cognitive ability.

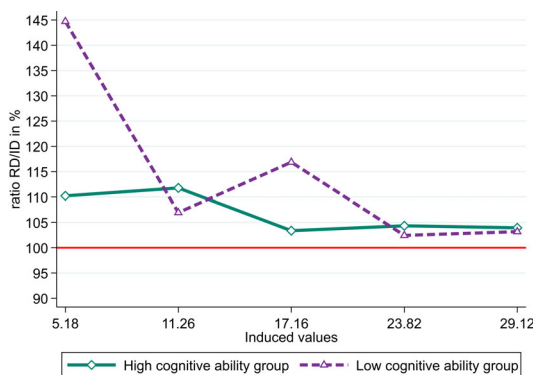


Figure 2. Aggregate behavior by induced values by cognitive ability group

Figure 2 shows the ratio of revealed demand to induced demand in both cognitive groups by induced value. The average demand revelation ratio across induced values in the higher cognitive group is about 106.73%, while the average demand revelation ratio in the lower cognitive group is about 114.83%. Moreover, the variation of the demand revelation ratio across induced values in the lower cognitive group is much higher than that in the higher cognitive group, indicating that subjects with greater cognitive ability are less sensitive to changes in values for the auctioned item; thus, aggregate behavioral bias is less pronounced for the group of subjects with greater cognitive ability.

From figure 2, it is also clear that overbidding is particularly pronounced for low ability subjects with a low induced value.²³ For both groups, the ratio of revealed demand to induced demand tends to converge to the perfect demand revelation as induced values increase, which is consistent with previous results (e.g., Drichoutis, Lusk, and Nayga 2015), indicating that individual bidders tend to perceive a greater incentive to bid optimally in a second price auction as their induced values increase. That is, bidders appear to take their bidding more seriously when they are more likely to actually win the auction. This could also be evidence of “spite” because low value bidders are unlikely to win the auction but may set the price someone else pays. The speed of convergence as induced values increase tends to be faster, however, in the high cognitive group.

²³ This overbidding mainly comes from relatively large overbids. About 76% of total overbids are above 10% of induced value when the induced value is 5.18 in the lower cognitive group.

Table 4. Average Marginal Effects of Cognitive Ability on Incidence of Bid Deviation

	All	High cognitive group	Low cognitive group
	(1)	(2)	(3)
RSPM	−0.08*** (0.03)	−0.19*** (0.06)	0.05 (0.04)
Induced value	0.002** (0.001)	0.003** (0.001)	0.001 (0.001)
Gender	−0.06 (0.04)	−0.05 (0.07)	−0.10* (0.06)
Age	−0.005 (0.006)	−0.004 (0.01)	−0.008** (0.004)
Employed	−0.08 (0.05)	−0.11 (0.08)	−0.04 (0.06)
Extraversion	0.01 (0.02)	−0.03 (0.04)	0.07* (0.04)
Agreeableness	−0.01 (0.02)	0.02 (0.04)	−0.06 (0.04)
Conscientiousness	0.01 (0.02)	0.002 (0.03)	−0.02 (0.05)
Emotional Stability	−0.03 (0.03)	−0.07* (0.04)	0.04 (0.04)
Openness	0.05* (0.02)	0.09** (0.04)	0.02 (0.03)
Grit	0.001 (0.03)	0.002 (0.04)	−0.01 (0.04)
CFC	−0.03 (0.02)	−0.04 (0.04)	−0.02 (0.03)
Round dummies	Yes	Yes	Yes
N	750	375	375

Note: Standard error for each coefficient shown in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%.

Cognitive Ability and Individual Bidding Behavior

In this section, we report regression results that examine whether cognitive ability influences the incidence of bid deviations in a SPA.²⁴ Because the two cognitive groups overlap in their scores, we look at the effect of the Raven test scores (RSPM) on the probability of bid deviation from induced value using a conditional Probit regression. In addition, given that we have individual bids for multiple auction rounds, we take into account the panel nature of our data by estimating random effects Probit models.

Column (1) of table 4 reports the marginal effect estimates from a random effects Probit regression with covariates that include demographics and personality trait.²⁵ The dependent variable is a dummy for bid deviation among all bids (bid deviation = 1, perfect demand revealing = 0). Raven scores and character skills are standardized. Results show that the RSPM is negatively related to the probability of bid deviation from induced value. That is, subjects with higher cognitive ability are less likely to deviate from induced value. We also examine the effect of cognitive ability on the incidence of bid deviation in each cognitive group. Results are reported in

columns (2) and (3) of table 4. We can see that cognitive ability is negatively related to the probability of bid deviation in the higher cognitive group, while the effect of cognitive ability is not statistically significant in the lower cognitive group. This result could suggest that one had to have cognitive ability above some level before one is able to see that bid deviation from induced value is a bad decision.

We next examine if cognitive ability has an impact on the incidence of bid deviations separately for each induced value and for each round with the same control variables as in table 4. Tables 5 and 6 present the marginal effect estimates from a Probit regression for each case.²⁶ For brevity, we only post the coefficient of the Raven test scores. Table 5 shows that subjects with higher test scores are less likely to deviate from induced value for all induced values. Table 6 also shows that test scores are negatively related to the probability of bid deviations from induced value in all bidding rounds.

Cognitive Ability and Overbidding and Underbidding Behavior

Next, we examine whether cognitive ability can explain the patterns of overbidding and underbidding behaviors in SPAs by classifying overbids and underbids into two categories: small overbid or small underbid (overbids or

²⁴ We also compare truthful bidding behavior between the two cognitive groups by estimating the bidding functions. The results are reported in the Supplementary Material E in the Appendix.

²⁵ The coefficients are available in Supplementary Material F in the Appendix. Our results are robust with and without the control variables. All results without controls are available in Supplementary Material G in the Appendix.

²⁶ Our results are robust with and without control variables. All results without controls are available in Supplementary Material G in the Appendix.

Table 5. Average Marginal Effects of Cognitive Ability on Incidence of Bid Deviation by Induced Value

	Induced value				
	5.18	11.26	17.16	23.82	29.12
RSPM	−0.10 *** (0.04)	−0.11 *** (0.04)	−0.06* (0.03)	−0.09** (0.03)	−0.09** (0.03)
N	150	150	150	150	150

Note: Standard error for each coefficient shown in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1% respectively.

Table 6. Average Marginal Effects of Cognitive Ability on Incidence of Bid Deviation by Rounds

	Bidding Round				
	Round 1	Round 2	Round 3	Round 4	Round 5
RSPM	−0.11 *** (0.03)	−0.09 *** (0.03)	−0.07* (0.04)	−0.10 *** (0.03)	−0.07** (0.03)
N	150	150	150	150	150

Note: Standard error for each coefficient shown in parentheses. * and *** denote statistical significance at 10% and 1%.

underbids $\leq 10\%$ of induced value) and large overbid or underbid (10% of induced value < overbids or underbids).²⁷

Figure 3 exhibits the percentage of each type of overbids among total overbids in both cognitive groups. From the figure, it is clear that overbidding has a different pattern between the two cognitive groups, indicating that the pattern of overbidding is not homogeneous across individuals, consistent with the finding from Cooper and Fang (2008). Subjects in the higher cognitive group submit more small overbids while subjects in the lower cognitive group submit more large overbids. The two-sample proportion tests also reject the null hypothesis of equal proportions between the two cognitive groups in both small overbid (z-statistic: 3.19, p-value: 0.001) and large overbid (z-statistic: −3.19, p-value: 0.001).

Figure 4 shows the percentage of each type of underbids among total underbids. Subjects with higher cognitive ability are less likely to submit small underbids, but they are more likely to submit large underbids. However, a proportion test fails to reject the hypothesis of equal proportions between the two cognitive groups in small underbids (z-statistic: −1.59, p-value: 0.11) and large underbids (z-statistic: 1.59, p-value: 0.11).

We also examine whether cognitive ability impacts the patterns of overbidding and

underbidding behavior using Probit models. The marginal effects are shown in table 7.²⁸ The dependent variable in model (1) is a dummy for large overbids among all overbids (large overbid = 1, small overbid = 0); in model 2, the dependent variable is a dummy for large underbids among all underbids (large underbid = 1, small underbid = 0). The results confirm our observation that subjects with greater cognitive ability are more likely to submit small overbids and less likely to submit large overbids as compared to subjects with lower cognitive ability. The coefficient of the Raven test scores is not statistically significant in the case of underbids, indicating that the patterns of underbidding are similar between more cognitively able subjects and less cognitively able subjects.

Experiment 2

Experimental Design

To test the robustness of the results from Experiment 1, we conducted a second experiment using subjects from the general population rather than college students.²⁹ Given that in Experiment 1 the assignment of

²⁷ The categories are somewhat arbitrary. We define a small overbid as one when bids are within 10% of induced value because subjects with small overbids have a less chance to win the auction and thus would not experience a negative result. We also use the same categories for the underbid case.

²⁸ Coefficients of the Probit regression are available in Supplementary Material F in the Appendix. Our results are robust with and without controls. The results without controls are available in Supplementary Material G in the Appendix.

²⁹ We are grateful to the anonymous reviewers for this suggestion.

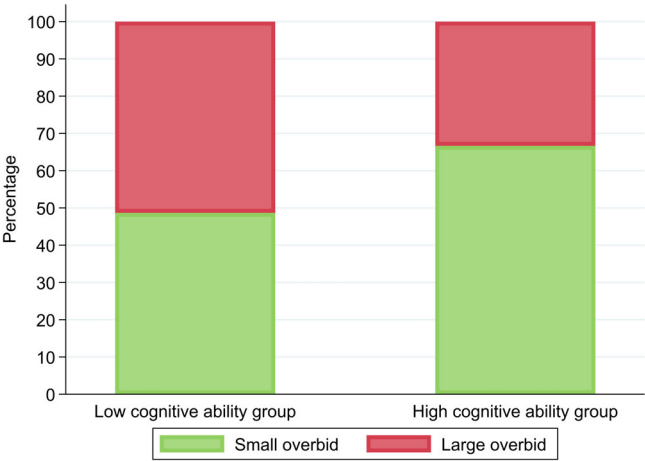


Figure 3. Pattern of overbidding in cognitive groups

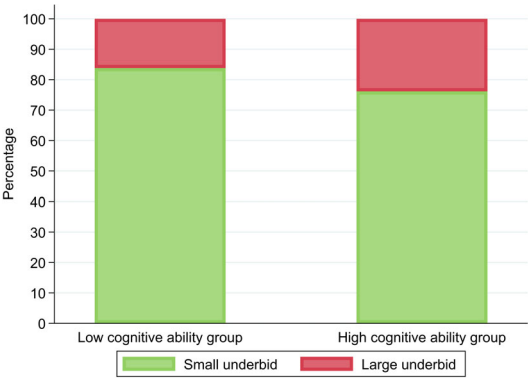


Figure 4. Pattern of underbidding in cognitive groups

subjects to the high or low ability group did not affect subjects’ bidding behaviors (See Supplementary Material B in the Appendix), in Experiment 2, we did not sort subjects into groups based on their cognitive ability. Instead, we randomly assigned subjects into groups consisting of five subjects.³⁰ This design change enables us to randomize the order of our main tasks (the RSPM and the SPA) to avoid potential order effect that could be present in Experiment 1. With the exception of these changes, our design was consistent with that of Experiment 1.³¹

³⁰ Given the finding of previous studies that the number of bidders can affect the equilibrium bid (Banerji and Gupta 2014; Rosato and Tymula 2019), we kept constant the number of subjects in a group across sessions.

³¹ Experiment 2 did not include the alternative measure of cognitive ability that was implemented in Experiment 1 (see Footnote 15).

We conducted a total of twelve new sessions. Each session included ten subjects, none of which had participated in Experiment 1. All sessions were conducted between April 2019 and May 2019 in the same lab used for Experiment 1. The subjects were recruited via paper and online advertisements distributed in the community. Subjects were paid \$5 if they showed up on time and they were paid an extra \$35 if they participated in the experiment.³² The characteristics of subjects are exhibited in table 8. The average RSPM test score for subjects from the general population is 49.3, which is similar to that for low cognitive group in our student subjects.

Experimental Results

Table 9 reflects the bidding behavior from the experiment with non-student subjects. On average, 18.3% of bids are perfectly demand revealing, 38.7% of bids are overbids, and 43% of bids are underbids. The percentage of overbids is smaller, while the percentage of underbids is greater in Experiment 2 (non-students) than in Experiment 1 (students). The proportion of large overbids (i.e., bids above 10% of induced value) is similar across

³² The higher participation payment was necessary to attract adults in the community who typically would have higher opportunity cost as well as greater transactions cost, such as parking. Laboratory studies are more convenient for students, which is why students are typically recruited for studies. The induced values remained the same as in Experiment 1 so that the results would be directly comparable. Any losses the subjects earned in the experiment were deducted from their participation payment. In the auctions, subjects were paid their salient earnings, which averaged \$4.80 (min = −18.61, max = 18.94).

Table 7. Average Marginal Effects of Cognitive Ability on the Patterns of Overbids and Underbids

	(1) Probability of submitting large overbid	(2) Probability of submitting large underbid
RSPM	−0.09** (0.04)	−0.03 (0.03)
Induced value	−0.01*** (0.002)	−0.001 (0.002)
Gender	0.11 (0.08)	0.005 (0.06)
Age	−0.03** (0.01)	−0.02** (0.01)
Employed	0.09 (0.08)	0.01 (0.06)
Extraversion	−0.07* (0.04)	−0.01 (0.03)
Agreeableness	0.04 (0.04)	0.07** (0.03)
Conscientiousness	0.02 (0.04)	−0.01 (0.04)
Emotional Stability	−0.04 (0.05)	0.04 (0.04)
Openness	0.07 (0.05)	−0.01 (0.03)
Grit	−0.05 (0.05)	−0.11*** (0.04)
CFC	0.05 (0.05)	0.06 (0.04)
Round dummies	Yes	Yes
N	308	282

Note: Standard error for each coefficient shown in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%.

Table 8. Subjects' Characteristics (non-student)

Variables	Mean (standard deviation)
RSPM	49.31 (8.15)
Age (years)	41.15 (14.37)
Gender (1: male, 0: female)	35.83%
Employed (1: yes, 0: no)	81.67%
Big5: Extraversion	8.51 (3.10)
Big5: Agreeableness	10.48 (2.51)
Big5: Conscientiousness	11.26 (2.31)
Big5: Emotional stability	10.18 (2.82)
Big5: Openness	10.79 (2.17)
Grit	43.25 (7.41)
CFC	44.12 (6.95)

Note: Each value represents the average of each variable. Standard deviation for each value is shown in parentheses. The Big Five personality traits use a seven-point Likert scale to measure responses to individual questions; the Grit scale and CFC use a five-point Likert scale to measure responses to each question.

the two experiments. Moreover, the percentage of large overbids in Experiment 2 decreases as their induced values increase, which is consistent with the finding in the experiment with student subjects. Non-student subjects are more likely to submit larger underbids than student subjects, and this bidding pattern does not vanish as induced values increase.

We mainly examine whether cognitive ability affects individual subjects' bidding behaviors. Specifically, we re-ran the Probit regression to

investigate the effect of cognitive ability on the incidence of bid deviations in the SPA. Furthermore, we re-ran the Probit regression to examine whether cognitive ability can explain the patterns of overbidding and underbidding in SPAs.

Column (1) of table 10 reports the results of the effect of cognitive ability on the incidence of bid deviations with the same covariates as in table 4³³. Results confirm our earlier finding that subjects with higher cognitive ability are less likely to deviate from induced value compared to subjects with lower cognitive ability. Results in columns 2 and 3 of table 10 also confirm our finding that subjects with higher cognitive ability are less likely to submit large overbids and large underbids compared with subjects with lower cognitive ability. However, for the case of overbids, the coefficient of the Raven test score is not statistically significant.

Conclusions

Researchers know that there are considerable behavioral anomalies in bidding behavior in experimental auctions. One of the consistent findings in previous experimental studies is that people deviate from rational behavior and often submit bids above their own

³³ The coefficients of the random effects Probit regression are available in Supplementary Material F in the Appendix.

Table 9. Frequency of Bids Relative to Induced Value (non-student)

		Number			Percentage		
		=	>	<	=	>	<
By round	R1	22	52 (29)	46 (25)	18.3	43.3 (55.8)	38.3 (54.3)
	R2	19	46 (26)	55 (26)	15.8	38.3 (56.5)	45.8 (47.3)
	R3	24	48 (26)	48 (20)	20.0	40.0 (54.2)	40.0 (41.7)
	R4	22	43 (24)	55 (27)	18.3	35.8 (55.8)	45.8 (49.1)
	R5	23	43 (24)	54 (28)	19.2	35.8 (55.8)	45.0 (51.9)
By value	5.18	28	44 (14)	48 (27)	23.3	36.7 (31.9)	40.0 (56.2)
	11.26	19	53 (25)	48 (21)	15.8	44.2 (47.2)	40.0 (43.8)
	17.16	19	46 (27)	55 (29)	15.8	38.3 (58.7)	45.8 (52.7)
	23.82	22	46 (31)	52 (22)	18.3	38.3 (67.4)	43.3 (42.3)
	29.12	22	43 (32)	55 (27)	18.3	35.8 (74.4)	45.8 (49.1)

Note: =, >, and < denote bid equal to induced value, bid larger than induced value, and bid smaller than induced value. Entries in parentheses are the number/percentage of individuals whose bids are within 10% of his/her induced value among subjects who overbid or underbid.

valuation in second price auctions. Deciphering the reasons for these behavioral anomalies helps to assess the appropriateness of using these mechanisms to infer people’s home-grown values.

We investigate how subjects’ bidding behaviors in the second price auction vary with their cognitive ability. This topic is particularly relevant given Li’s (2017) claim that second price auctions are not obviously strategy proof. Our study is also motivated by previous findings about the importance of understanding an auction mechanism (e.g., Kagel, Harstad, and Levin 1987; Ausubel 2004; Li 2017) and the role of cognitive ability in decision making and strategy games (e.g., Burnham et al. 2009; Dohmen et al. 2010; Smith, McArdle, and

Willis 2010; Carpenter, Graham, and Wolf 2013; Gill and Prowse 2016). To examine our objectives, in Experiment 1 we gave each (student) subject a cognitive ability test, then classified subjects based on the outcome of the test and matched them in auction groups in order to participate in a series of second price auctions.

From our first experiment (with student subjects), we found that at the aggregate level, the pattern of overbidding is somewhat moderated for subjects with greater cognitive ability. At the individual level, subjects with greater cognitive ability tended to submit bids closer to their induced values than subjects with lower cognitive ability. We also found that overbidding and bid deviation are more

Table 10. Average Marginal Effects of Cognitive Ability on Incidence of Bid Deviation (non-student sample)

	(1) Probability of bid deviation	(2) Probability of large overbid	(3) Probability of large underbid
RSPM	−0.08** (0.04)	−0.06 (0.06)	−0.19*** (0.05)
Induced value	0.001 (0.001)	−0.02*** (0.002)	0.003 (0.002)
Gender	−0.11** (0.05)	−0.12 (0.10)	−0.12 (0.09)
Age	0.001 (0.002)	−0.006 (0.004)	−0.004 (0.004)
Employed	−0.02 (0.06)	0.05 (0.11)	−0.10 (0.13)
Extraversion	−0.00 (0.02)	0.02 (0.05)	0.04 (0.04)
Agreeableness	−0.03 (0.03)	−0.08* (0.05)	−0.10** (0.05)
Conscientiousness	−0.001 (0.02)	−0.03 (0.04)	−0.04 (0.06)
Emotional Stability	−0.02 (0.03)	0.07 (0.06)	−0.07 (0.05)
Openness	−0.01 (0.03)	−0.005 (0.05)	−0.006 (0.05)
Grit	0.04 (0.03)	0.01 (0.05)	0.06 (0.06)
CFC	−0.04 (0.03)	−0.04 (0.05)	0.04 (0.05)
Round dummies	Yes	Yes	Yes
N	600	232	258

Note: Standard errors are shown in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%.

pronounced for subjects with lower cognitive ability when values for the item are relatively small (i.e., small induced values). Regardless of cognitive ability, underbidding behavior is similar. Our findings provide evidence of the importance of cognitive ability in explaining bid deviation and overbidding behavior in SPAs. An interesting extension of our study would be an investigation about why subjects with lower cognitive ability are more likely to submit large overbid.

We also conducted an additional experiment using subjects drawn from the general population to check the external validity of our findings. Results from this additional experiment confirmed that subjects with greater cognitive ability are less likely to deviate from induced values. In addition, subjects with higher cognitive ability are less likely to submit large underbids than subjects with lower cognitive ability.

Overall, our results provide important evidence that more cognitively able subjects behave in closer accordance with theory and that cognitive skills can partially explain heterogeneity in bidding behavior in SPAs. Our findings also support the notion that the difficulty of understanding the SPA could lead to irrational bidding behavior and that the dominant strategy in SPAs tends to be much less apparent to cognitively limited subjects.

Second, our results have important implications for future studies using second price auctions to infer homegrown values. Specifically, researchers should collect a measure of cognitive ability when performing such studies to better identify the true underlying demand conditions by being able to discount data collected from low ability subjects. They could also use this measure as a control variable in the conditional analysis or use this to check for heterogeneity across cognitive ability of subjects. Previous studies have developed and validated short forms to measure people's cognitive ability (e.g. Bors and Stokes 1998; Bilker et al. 2012). Researchers may use these simple forms of cognitive measure to collect subjects' cognitive ability in applications of SPAs in market settings. For researchers who are hesitant to collect cognitive ability, our findings could indirectly and partly imply that it is really important to spend time making sure that subjects clearly understand the SPA mechanism, given the propensity of subjects, especially those who have lower cognitive ability, to misbid in SPAs. Previous studies argued that clearly understanding an auction

mechanism is important to minimize behavioral anomalies and make auction methods a valid measurement instrument (see Kagel, Harstad, and Levin 1987; Plott and Zeiler 2005). Given our findings pertaining to greater incidence of overbidding with low induced values, this should also be kept in mind by researchers conducting homegrown SPAs for low value products. Specifically, aggregate values are closer to perfect demand revelation regardless of cognitive ability as the induced values for the item increases. This suggests that the second price auction could be more useful for higher valued products than lower valued products in homegrown valuation studies.

Supplementary Material

Supplementary material are available at *American Journal of Agricultural Economics* online.

References

- Agarwal, Sumit, and Bhashkar Mazumder. 2013. Cognitive abilities and household financial decision making. *American Economic Journal: Applied Economics* 5(1): 193–207.
- Andersson, Henrik, and Mikael Svensson. 2008. Cognitive ability and scale bias in the contingent valuation method. *Environmental and Resource Economics* 39 (4): 481–95.
- Andreoni, James, Yeon-Koo Che, and Jinwoo Kim. 2007. Asymmetric information about rivals' types in standard auctions: An experiment. *Games and Economic Behavior* 59(2): 240–59.
- Ausubel, Lawrence M. 2004. An efficient ascending-bid auction for multiple objects. *American Economic Review* 94 (5): 1452–75.
- Ballinger, Parker T, Eric Hudson, Leonie Karkoviata, and Nathaniel T Wilcox. 2011. Saving behavior and cognitive abilities. *Experimental Economics* 14(3): 349–74.
- Banerji, A, and Neha Gupta. 2014. Detection, identification, and estimation of loss aversion: Evidence from an auction experiment. *American Economic Journal: Microeconomics* 6(1): 91–133.
- Benjamin, Daniel J., Sebastian A. Brown., and Jesse M. Shapiro. 2013. Who is 'behavioral'?

- Cognitive ability and anomalous preferences. *Journal of European Economic Association* 11(6): 1231–1255.
- Bergman, Oscar, Tore Ellingsen, Magnus Johannesson, and Cicek Svensson. 2010. Anchoring and cognitive ability. *Economics Letters* 107(1): 66–8.
- Bilker, Warren B, John A Hansen, Colleen M Brensinger, Jan Richard, Raquel E Gur, and Ruben C Gur. 2012. Development of abbreviated nine-item forms of the Raven's standard progressive matrices test. *Assessment* 19(3): 354–69.
- Borghans, Lex, Huub Meijers, and Bas Ter Weel. 2008. The role of noncognitive skills in explaining cognitive test scores. *Economic Inquiry* 46(1): 2–12.
- Bors, Douglas A, and Tonya L Stokes. 1998. Raven's Advanced Progressive Matrices: Norms for first-year university students and the development of a short form. *Educational and Psychological Measurement* 58(3): 382–98.
- Brañas-Garza, Pablo, Teresa Garcia-Muñoz, and Rober H González. 2012. Cognitive effort in the beauty contest game. *Journal of Economic Behavior & Organization* 83(2): 254–60.
- Burnham, Terence C, David Cesarini, Magnus Johannesson, Paul Lichtenstein, and Björn Wallace. 2009. Higher cognitive ability is associated with lower entries in a p-beauty contest. *Journal of Economic Behavior & Organization* 72(1): 171–5.
- Canavari, Maurizio, Andreas C Drichoutis, Jayson L Lusk, and Rodolfo M Nayga, Jr. 2019. How to run an experimental auction: A review of recent advances. *European Review of Agricultural Economics* 46(5): 862–922.
- Carpenter, Jeffrey, Michael Graham, and Jesse Wolf. 2013. Cognitive ability and strategic sophistication. *Games and Economic Behavior* 80: 115–30.
- Chen, Chia-Ching, I-Ming Chiu, John Smith, and Tetsuji Yamada. 2013. Too smart to be selfish? Measures of cognitive ability, social preferences, and consistency. *Journal of Economic Behavior & Organization* 90: 112–22.
- Cherry, Todd, Peter Frykblom, Jason Shogren, John List, and Melonie Sullivan. 2004. Laboratory testbeds and non-market valuation: The case of bidding behavior in a second-price auction with an outside option. *Environmental and Resource Economics* 29(3): 285–94.
- Civelli, Andrea, and Cary Deck. 2018. A flexible and customizable method for assessing cognitive abilities. *Review of Behavioral Economics* 5(2): 123–47.
- Cooper, David J, and Hanming Fang. 2008. Understanding overbidding in second price auctions: An experimental study. *Economic Journal* 118(532): 1572–95.
- Corrigan, Jay R, Andreas C Drichoutis, Jayson L Lusk, Rodolfo M Nayga, and Matthew C Rousu. 2012. Repeated rounds with price feedback in experimental auction valuation: An adversarial collaboration. *American Journal of Agricultural Economics* 94(1): 97–115.
- Dohmen, Thomas, Armin Falk, David Huffman, and Uwe Sunde. 2010. Are risk aversion and impatience related to cognitive ability? *American Economic Review* 100(3): 1238–60.
- Drichoutis, Andreas C, Stathis Klonaris, and Georgia S Papoutsis. 2017. Do good things come in small packages? Bottle size effects on willingness to pay for pomegranate wine and grape wine. *Journal of Wine Economics* 12(1): 84–104.
- Drichoutis, Andreas C, Jayson L Lusk, and Rodolfo M Nayga. 2015. The veil of experimental currency units in second price auctions. *Journal of Economic Science Association* 1(2): 182–96.
- Duckworth, Angela L, Christopher Peterson, Michael D Matthews, and Dennis R Kelly. 2007. Grit: Perseverance and passion for long-term goals. *Journal of Personality and Social Psychology* 92(6): 1087–101.
- Edwards, Ward. 1968. Conservatism in human information processing. In *Formal Representation of Human Judgement*, ed. Benjamin Kleinmütz, 17–52. New York: Wiley.
- Fischbacher, Urs. 2007. z-Tree: Zurich toolbox for ready-made economic experiments. *Experimental Economics* 10(2): 171–8.
- Frederick, Shane. 2005. Cognitive reflection and decision making. *Journal of Economic Perspectives* 19(4): 25–42.
- Frey, Meredith C, and Douglas K Detterman. 2004. Scholastic assessment or g? The relationship between the scholastic assessment test and general cognitive ability. *Psychological Science* 15(6): 373–8.
- Georganas, Sotiris, Dan Levin, and Peter McGee. 2017. Optimistic irrationality and overbidding in private value auctions. *Experimental Economics* 20(4): 772–92.

- Gignac, Gilles E. 2018. A moderate financial incentive can increase effort, but not intelligence test performance in adult volunteers. *British Journal of Psychology* 109 (3): 500–16.
- Gill, David, and Victoria Prowse. 2016. Cognitive ability, character skills, and learning to play equilibrium: A level-k analysis. *Journal of Political Economy* 124(6): 1619–76.
- Gosling, Samuel D, Peter J Rentfrow, and William B Swann. 2003. A very brief measure of the Big-Five personality domains. *Journal of Research in Personality* 37(6): 504–28.
- Gray, Jeremy R, and Paul M Thompson. 2004. Neurobiology of intelligence: Science and ethics. *Nature Reviews Neuroscience* 5(6): 471–82.
- Grebitus, Carola, Jayson Lusk, and Rodolfo M Nayga, Jr. 2013. Explaining differences in real and hypothetical experimental auctions and choice experiments with personality. *Journal of Economic Psychology* 36: 11–26.
- Harrison, Glenn W. 1989. Theory and misbehavior of first-price auctions. *American Economic Review* 79(4): 749–62.
- . 1992. Theory and misbehavior of first-price auctions: Reply. *American Economic Review* 82(5): 1426–43.
- Harstad, Ronald M. 2000. Dominant strategy adoption and bidders' experience with pricing rules. *Experimental Economics* 3 (3): 261–80.
- Hoppe, Eva I, and David J Kusterer. 2011. Behavioral biases and cognitive reflection. *Economics Letters* 110(2): 97–100.
- Jacquemet, Nicolas, Robert-Vincent Joule, Stéphane Luchini, and Jason F Shogren. 2013. Preference elicitation under oath. *Journal of Environmental Economics and Management* 65(1): 110–32.
- Jensen, Arthur R. 1998. *The g factor: The Science of Mental Ability*. Westport, CT: Praeger.
- Jones, Matthew T. 2014. Strategic complexity and cooperation: An experimental study. *Journal of Economic Behavior & Organization* 106: 352–66.
- Kagel, John H, Ronald M Harstad, and Dan Levin. 1987. Information impact and allocation rules in auctions with affiliated private values: A laboratory study. *Econometrica: Journal of Econometric Society* 55(6): 1275–304.
- Kagel, John H, and Dan Levin. 1993. Independent private value auctions: Bidder behaviour in first-, second-and third-price auctions with varying numbers of bidders. *Economic Journal* 103(419): 868–79.
- Kahneman, Daniel, and Amos Tversky. 1982. *Evidential impact of base rates. Judgment under uncertainty: Heuristics and biases*. Cambridge, UK: Cambridge University Press.
- Li, Shengwu. 2017. Obviously strategy-proof mechanisms. *American Economic Review* 107(11): 3257–87.
- Lusk, Jayson L, Corinne Alexander, and Matthew C Rousu. 2007. Designing experimental auctions for marketing research: The effect of values, distributions, and mechanisms on incentives for truthful bidding. *Review of Marketing Science* 5(1): 1–32.
- Morgan, John, Ken Steiglitz, and George Reis. 2003. The spite motive and equilibrium behavior in auctions. *Contributions in Economic Analysis & Policy* 2(1): 1–25.
- Oechssler, Jörg, Andreas Roider, and Patrick W Schmitz. 2009. Cognitive abilities and behavioral biases. *Journal of Economic Behavior & Organization* 72(1): 147–52.
- Plott, Charles R, and Kathryn Zeiler. 2005. The willingness to pay-willingness to accept gap, the "endowment effect," subject misconceptions, and experimental procedures for eliciting valuations. *American Economic Review* 95(3): 530–45.
- Rosato, Antonio, and Agnieszka A Tymula. 2019. Loss aversion and competition in Vickrey auctions: Money ain't no good. *Games and Economic Behavior* 115: 188–208.
- Rutström, Elisabet E. 1998. Home-grown values and incentive compatible auction design. *International Journal of Game Theory* 27(3): 427–41.
- Slonim, Robert, James Carlson, and Eric Bettinger. 2007. Possession and discounting behavior. *Economics Letters* 97(3): 215–21.
- Smith, James P, John J McArdle, and Robert Willis. 2010. Financial decision making and cognition in a family context. *Economic Journal* 120(548): F363–80.
- Strathman, Alan, Faith Gleicher, David S Boninger, and C Scott Edwards. 1994. The consideration of future consequences: Weighing immediate and distant outcomes of behavior. *Journal of Personality and Social Psychology* 66(4): 742–52.
- Tversky, Amos, and Daniel Kahneman. 1983. Extensional versus intuitive reasoning: The conjunction fallacy in probability judgment. *Psychological Review* 90(4): 293–315.